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Microbial electrolysis cell simultaneously enhancing methanization and reducing hydrogen sulfide production in anaerobic digestion of sewage sludge

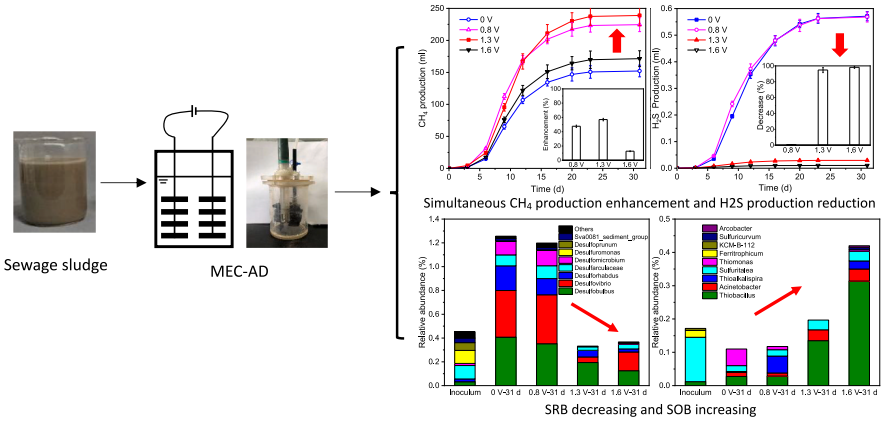
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HIGHLIGHTS

- The effects of high applied voltages on MEC-AD of sewage sludge were studied.
- Methanization enhancement and H_2S production reduction was simultaneously realized.
- High voltages (1.3 V and 1.6 V) created micro-aerobic condition for the AD.
- High voltages increased the abundances of SOB and decreased those of SRB.
- High voltages enhanced the hydrogenotrophic methanogenesis.

GRAPHICAL ABSTRACT



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ABSTRACT

The effects of microbial electrolysis cells (MECs) at three applied voltages (0.8, 1.3, and 1.6 V) on simultaneously enhancing methanization and reducing hydrogen sulfide (H_2S) production in the anaerobic digestion (AD) of sewage sludge were studied. The results showed that the MECs at 1.3 V and 1.6 V simultaneously enhanced the methane production by 57.02 and 12.70% and organic matter removal by 38.77 and 11.13%, and reduced H_2S production by 94.8 and 98.2%, respectively. MECs at 1.3 V and 1.6 V created a micro-aerobic conditions for the digesters with oxidation-reduction potential as −178~-232 mv, which enhanced methanization and reduced H_2S production. Sulfur reduction, H_2S and elemental sulfur oxidation occurred simultaneously in the ADs at 1.3 V and 1.6 V. The relative abundances of sulfur-oxidizing bacteria increased from 0.11% to 0.42% and those of sulfur-reducing bacteria decreased from 1.24% to 0.33% when the applied voltage of MEC increased from 0 V to 1.6 V. Hydrogen produced by electrolysis enhanced the abundance of *Methanobacterium* and changed the methanogenesis pathway.

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1. Introduction

Sewage sludge, an inevitable by-product of biological wastewater treatment, is hazardous and contains various toxic substances such as pathogens, heavy metals, and organic contaminants (Wu et al., 2020). These contaminants may cause significant pollution and human health risks without appropriate treatment and disposal (Xiao et al., 2020; Yakamercan et al., 2021). Anaerobic digestion (AD) is a traditional method for treatment of sewage sludge and has been widely applied in wastewater treatment plants (WWTPs) worldwide (Di Capua et al., 2020; Xiao et al., 2020). AD converts part of the organic matter in the sludge into biogas containing 60–70% methane, which is a promising renewable energy source. The biogas energy potential is one of the most attractive advantages of AD for minimizing sludge from WWTPs (Díaz et al., 2010).

Although methane and carbon dioxide are its main constituents, biogas also contains significant quantities of undesirable compounds, such as hydrogen sulfide (H_2S) whose concentration can reach 1.0% (v/v) (Díaz et al., 2010; Krayzelova et al., 2015; Chen et al., 2020; Nguyen and Khanal, 2018). H_2S is formed by the reduction of inorganic (such as sulfate) and organic sulfur-containing compounds (such as proteins) in AD, and is a harmful substance in biogas and a vexing problem for the AD of sewage sludge (Krayzelova et al., 2015). H_2S must be removed before biogas is used as its concentration in biogas often exceeds the technical limits of energy technologies. H_2S also causes corrosion, toxicity, and odor-associated problems. In the liquid phase, dissolved H_2S can easily diffuse through the cell membrane into the cytoplasm, damaging methanogens, particularly acetotrophic methanogens (Krayzelova et al., 2015; Chen et al., 2020). In addition, H_2S-reducing bacteria, namely, sulfur-reducing bacteria (SRB), have a higher growth rate, higher substrate affinity for acetate and hydrogen (H_2), and outcompete methanogens, thereby resulting in low methane yield and high H_2S content in biogas (Liu et al., 2018; Nguyen and Khanal, 2018). Therefore, the presence of H_2S and SRB in AD limit methane production. It is necessary to control SRB to reduce H_2S production during the AD of sludge.

Microaeration, that is, supplying small quantities of air or oxygen, is an effective method to scavenge H_2S from biogas by oxidizing sulfide to elemental sulfur, riching sulfur oxidizing bacteria (SOBs), and inhibiting the growth of SRBs, except for *Desulfovibrio aerotolerans* (Díaz et al., 2010; Krayzelova et al., 2015; Chen et al., 2020; Liu et al., 2018). Microaeration is regarded as a highly efficient, simple, and economically feasible technology for removing H_2S from biogas (Díaz et al., 2010; Krayzelova et al., 2015). Moreover, microaeration can improve AD of sludge by promoting hydrolysis and acidification, preventing volatile fatty acids (VFAs) accumulation, increasing organic matter removal, enhancing biogas production, and stabilizing the overall digestion processs (Chen et al., 2020). In microaeration, hydrolytic and acidifying bacteria will consume the oxygen, decrease the oxidation-reduction potential around the methanogens, which protect the methanogens. Therefore, although methanogenesis is strictly anaerobic, the increasing oxidation-reduction potential in microaerobic condition created by microaeration can enhance methanization and increase methane production (Fu et al., 2023). Microaeration can be performed by air/oxygen injection, electrolytic aeration, and natural aeration (leaving the tested configuration open with or without shaking) (Girotto et al., 2018; Krayzelova et al., 2015). Although air/oxygen injection is the most studied method, electrolytic aeration has unique advantages: the oxygen produced by electrolysis can create microaeration conditions, while the hydrogen produced by electrolysis can improve the combustion characteristics of biogas and further increase the net production of methane (Tartakovsky et al., 2011). Additionally, electrolytic aeration can overcome a drawback of air injection, namely, the dilution of biogas with nitrogen (Girotto et al., 2018).

Microbial electrolysis cells (MEC) are a type of electrolysis cell that use electrochemically active bacteria to convert organic matter into hydrogen or a variety of by-products without polluting the environment at lower applied voltages (Hua et al., 2019). Recently, MEC-assisted AD (MEC-AD) has been investigated as a means of improving AD and increasing methane recovery (Yu et al., 2018; Wang et al., 2022). MEC-AD can accelerate the degradation of a substrate (including recalcitrant compounds) and alter the AD microbial community by enriching exoelectrogens and methanogens. In MEC-AD, exoelectrogens, such as *Geobacter sulfurreducens* and *Shewanella oneidensis*, anaerobically ferment organic matter to produce electrons, which is transferred to methanogens through direct electron transfer or direct interspecies electron transfer (DIET). When MEC was combined with AD, the suitable applied voltage of MEC can provide energy required for respiration of exoelectrogens and methanogens, enrich them, increase their activities, enhance DIET between them, conquer thermodynamics barriers therefore enhancing the organic acid degradation, promote H_2 formation to enhance hydrogenoetrophic methanogenesis (Wang et al., 2022; Yu et al., 2018). Thereby, the AD was enhanced and methane production increased. Despite the theoretically minimum applied voltage of 0.14 V, most MEC-AD were operated at applied voltages of 0.3–1.5 V (Yu et al., 2018; Wang et al., 2022). Applied voltages above 1.0 V (high voltage) are generally not recommended because the electrical energy input is so large that the microbial electrolysis process becomes similar to a water electrolysis process and may show negative effects (Yu et al., 2018; Wang et al., 2022). However, some studies have shown that appropriately high applied voltages (>1.0 V) can also improve AD by creating microaerobic conditions through in situ electrolytic oxygen (Chen et al., 2016; Ding et al., 2021; Tartakovsky et al., 2011; Xiao et al., 2018). For example, Tartakovsky et al. (2011) studied the effect of an applied voltage of 2.8–3.5 V on the AD of synthetic wastewater in MEC-up-flow anaerobic sludge bed and found that oxygen and hydrogen were produced by electrolysis, which created microaerobic conditions and enhanced methane production by 10–25%. Chen et al. (2016) found that the applied voltage of 1.2 V and 1.5 V increased the methane yield of waste-activated sludge AD by 40.5–51.6%, although this was lower than that of 0.6 V (76.2%).

Since MEC-AD, at a suitable applied voltage can create microaerobic conditions, it is expected that it can reduce the H_2S concentration in biogas. However, the effects of applied voltage on simultaneous methanization enhancement and H_2S production reduction for MEC-AD have not been investigated, and the resultant environmental benefits have been neglected. Therefore, the objective of this study was to study the effect of applied voltage on methanization and H_2S production in the MEC-AD of sewage sludge.

2. Material and methods

2.1. Sewage sludge and inoculum

Sewage sludge used in the test was obtained from a WWTP in Beijing, China. Anaerobic digestate from an anaerobic digester in the same WWTP was used as the inoculum for AD. The characteristics of the sewage sludge and inoculum are summarized in Table 1.

2.2. MEC-assisted anaerobic digesters

The MEC-assisted anaerobic digesters were cylindrical plexiglass jars with two electrodes (Fig. S1). The volume of the digesters was 250 mL with a working volume of 180 mL. The electrodes were made of a cylindrical carbon rod (length × diameter = 8 × 0.5 cm^2) and four rectangular carbon felts (length × width × thickness = 2 × 1 × 0.5 cm^3), with the carbon rod passing through the center of the carbon felts. The distance between the two adjacent carbon felts was 0.5 cm. The digesters had three ports for outgassing, liquid sampling, and oxidation-reduction potential (ORP) electrode.

Table 1
Characteristics of sewage sludge and inoculum.

Item	Sewage sludge	Inoculum
pH	6.92 ± 0.12	7.33 ± 0.05
TS (g/L)	27.01 ± 0.21	8.11 ± 0.09
VS (g/L)	17.58 ± 0.18	4.26 ± 0.25
TCOD (mg/L)	18720 ± 110	5530 ± 120
SCOD (mg/L)	45 ± 13	952 ± 83

2.3. Anaerobic digestion of sewage sludge

Sewage sludge (120 mL) and the inoculum (60 mL) were added to each digester. The digesters were purged with nitrogen for 5 min to create an anaerobic environment prior to sealing. The digesters were then placed in a water bath shaker (120–130 RPM) and incubated at 37 ± 1 °C. The two electrodes of the digesters were connected to DC power supplies using wires. The voltages applied between the two electrodes were 0, 0.8, 1.2, and 1.6 V. The tests were conducted in triplicate, and the results are presented as the mean value.

2.4. Analytical methods

The biogas produced from the digesters was periodically measured by displacing a saturated aqueous NaCl solution (Guo et al., 2014). The methane content in the produced biogas was measured using a gas chromatograph (Agilent 6890 N, USA) equipped with a flame ionization detector and a 30 m capillary column (HP-5). The H_2S content in the produced biogas was measured using another gas chromatograph (Agilent 6890, USA) equipped with a flame photometric detector and a 30 m capillary column (DB-1701).

Digestate pH was measured using a pH meter (Sartorius PB-10, Germany). The ORP of the digestate was measured in situ using a pH/ORP meter (KP-10Z, Kasahara Chemical Instruments Co., Japan). The sulfate concentration in the digestate was measured by ion chromatography (ICS-1000, Dionex, USA). The elemental sulfur on the electrodes was qualitatively analyzed using an elemental analyzer (Vario EL III, Germany). The soluble chemical oxygen demand (SCOD), total chemical oxygen demand (TCOD), soluble deoxyribonucleic acid (DNA), total solids, and volatile solids (VS) of the sewage sludge, inoculum, and digestate were measured using standard methods (APHA, 2012). The scanning electron microscopy (SEM) of digestates at the end of AD was conducted using a scanning electron microscope (JEOL JSM-7610 F, Japan).

2.5. Microbial analysis

According to a previous study (Xiao et al., 2020), the digestates sampled from the digesters at the end of the test were used to analyze their microbial communities. The DNA extraction of digestates, polymerase chain reaction (PCR) amplification of the extracted DNA, and Illumina MiSeq sequencing of the amplicons were performed following the methodology used in a previous study (Xiao et al., 2020), and the detailed processes are summarized in Supplementary Material S1. The raw reads were deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive database (accession number: PRJNA883421). The sequencing data was processed as described in

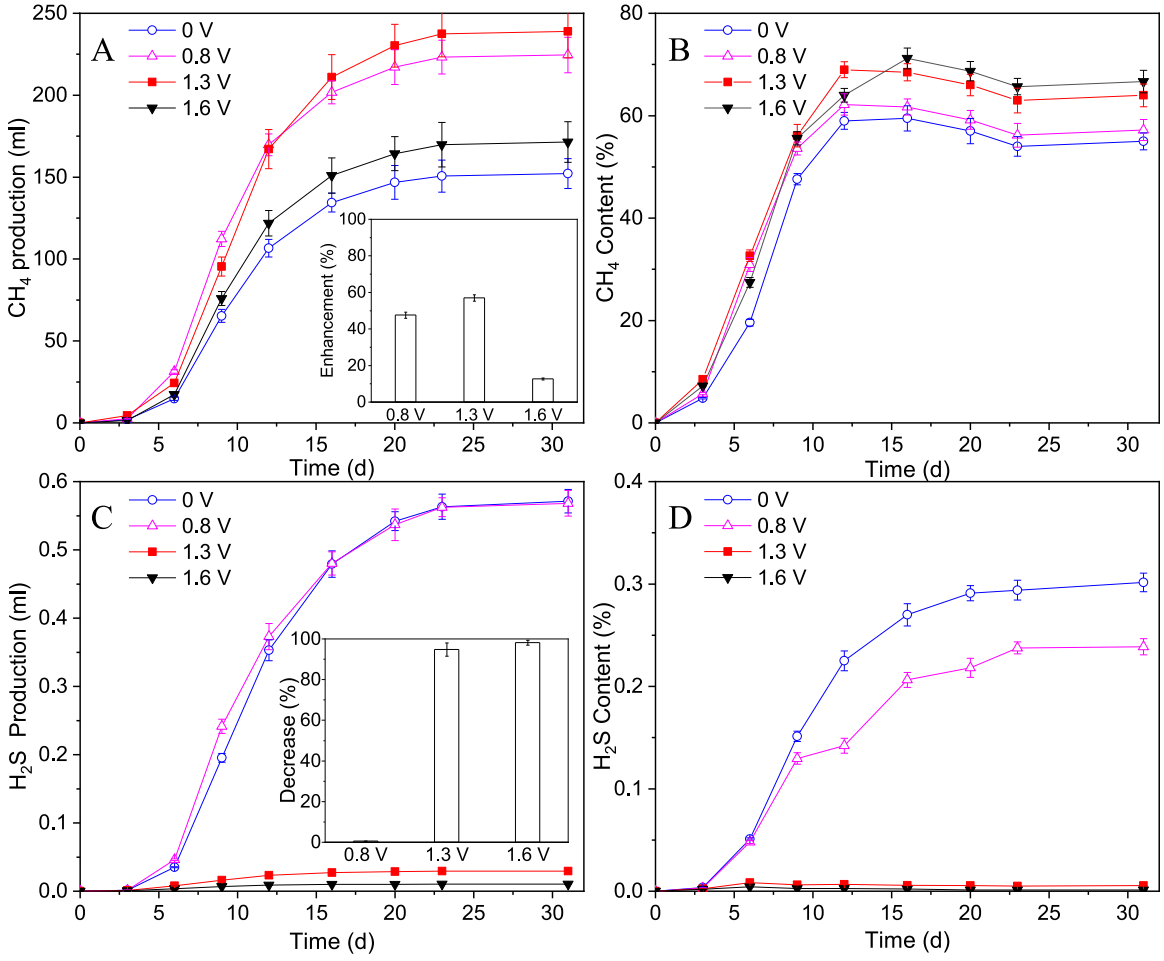


Fig. 1. Production and content of CH_4 and H_2S in the test.

Table 2
Matter removals of the four digesters at the end of the test.

Item	Start sample	End sample			
		0 V	0.8 V	1.3 V	1.6 V
TS (g/L)	20.82 ± 1.14	13.31 ± 0.25	11.14 ± 0.52	10.75 ± 0.22	12.60 ± 0.44
Removal of TS (%)	–	36.07 ± 3.15	46.49 ± 0.41	48.51 ± 2.23	39.48 ± 1.35
Enhancement (%)	–	–	28.89 ± 2.25	34.49 ± 1.32	9.45 ± 1.78
VS (g/L)	13.24 ± 0.85	8.21 ± 0.34	6.62 ± 1.41	6.36 ± 0.28	7.65 ± 0.46
Removal of VS (%)	–	37.99 ± 2.12	50.00 ± 1.29	52.72 ± 3.25	42.22 ± 4.36
Enhancement (%)	–	–	31.61 ± 2.65	38.77 ± 3.21	11.13 ± 0.55
TCOD (mg/L)	14340 ± 190	10110 ± 250	8170 ± 100	7880 ± 130	9440 ± 120
Removal of TCOD (%)	–	29.50 ± 2.24	43.03 ± 2.38	45.05 ± 3.13	34.17 ± 0.27
Enhancement (%)	–	–	47.39 ± 3.54	54.24 ± 2.54	17.36 ± 2.23

previous studies (Ni et al., 2018; Xiao et al., 2020). Aligned sequences were grouped into universal operational taxonomic units (OTUs) at a threshold of 97% minimum similarity, which has been widely accepted as the closest approximation for a microbial "species," short of culture-dependent analyses (Nguyen et al., 2016; Ni et al., 2018). Individual sequences were also grouped into OTUs to identify microbial distribution in the digestates. According to the biochemical characteristics of various methanogens (based on genus level) in the four MEC-ADs and main the substrate for methane production, the ratio of various methanogenesis pathways were calculated (Zheng et al., 2021).

3. Results and discussion

3.1. Performances of anaerobic digestion

The methane production of the four MEC-ADs is summarized in Fig. 1-A and B. At the end of the test, the cumulative methane production of the four MEC-ADs was 151 ± 10 mL (0 V), 223 ± 15 mL (0.8 V), 237 ± 13 mL (1.3 V), and 170 ± 14 mL (1.6 V) (Fig. 1-A). The three applied voltages enhanced the methane production of sludge by 47.62 ± 1.67% (0.8 V), 57.02 ± 1.71% (1.3 V), and 12.70 ± 0.55% (1.6 V) at the end of the test (Fig. 1-A). The enhancement in methane production in this study was similar to that reported in previous studies (Ding et al., 2016; Xiao et al., 2018). The results showed that the highest methane production enhancement was achieved with an applied voltage of 1.3 V and the second highest with an applied voltage of 0.8 V. Our previous study (Xiao et al., 2018) also found that the enhancement effect at 0.8 V was lower than that at 1.3 V. The change in soluble DNA of digestate in the four MEC-ADs (Fig. S2) showed that the applied voltage as 1.6 V cause certain microbial damage (Xiao et al., 2015). Meanwhile, the SEM of digestate at the end of AD showed that the surface of microorganisms was rougher at the applied voltage of 1.6 V than others (Fig. S3), also indicating microbial damage occurrence. Therefore, the methane production of MEC-AD with 1.6 V was lower than those with 0.8 V and 1.3 V was owing to 1.6 V causing a certain microbial damage and lowering the microbial activity (including bacteria and archaea) (Chen et al., 2016). The applied voltages also increased the methane content of biogas, with 1.6 V having the highest increase (Fig. 1-B). Water or other matter electrolysis might occur in the digesters at 1.3 V and 1.6 V to produce hydrogen and oxygen because they exceeded the lowest water electrolysis voltage (Chen et al., 2016). Hydrogen would be produced in the digesters at 1.3 V and 1.6 V with the digester at 1.6 V producing more. The produced hydrogen was converted into methane by hydrogenotrophic methanogens, which decreased the content of CO2 and increased that of methane. However, no oxygen or hydrogen was detected in the biogas, suggesting that they were rapidly consumed by the microorganisms (Bao et al., 2020; Guo et al., 2021). Some aerobic and facultative anaerobic microorganisms can grow in AD (Toerien, 1967), and use oxygen produced by water electrolysis. Additionally, many microorganisms can use hydrogen in MEC-AD, which explains the absence of hydrogen (Guo et al., 2013; Hassanein et al., 2017).

In addition to producing methane as energy, another important function of AD is organic matter removal. The removal of sludge VS and TCOD for the four MEC-ADs at the end of the test are summarized in Table 2. The VS and TCOD removal for the four MEC-ADs ranged from 37.99% to 52.72% and 29.50%–45.05%, respectively (Table 2). The digester at the 1.3 V applied voltage had the highest organic matter removal and that at 0 V had the lowest. The results showed that the applied voltage increased the organic matter removal for the AD of sewage sludge by (11.13–38.77%), and the increase was consistent with their cumulative methane production (Fig. 1-A). This was logical because the removed organic matter was mainly converted into methane during AD (Xiao et al., 2018; Xu et al., 2020).

3.2. Production of hydrogen sulfide

H_2S is a harmful component of biogas produced in sludge AD (Krayzelova et al., 2015) (Fig. 1-C and D). The H_2S content of biogas for the four MEC-ADs at the end of the test was 0.302 ± 0.009% (0 V), 0.239 ± 0.008% (0.8 V), 0.005 ± 0.000% (1.3 V), and 0.001 ± 0.000% (1.6 V) (Fig. 1-D), which suggests that the applied voltages decreased the H_2S content of biogas. At the end of the test, the H_2S produced in MEC-AD at 0.8 V (0.568 ± 0.019 mL) was slightly lower than that at 0 V (0.571 ± 0.017 mL) (Fig. 1-C). H_2S produced in the MEC-ADs at 1.3 V (0.030 ± 0.001 mL) and 1.6 V (0.011 ± 0.000 mL) (Fig. 1-C) was far lower than the production at 0 V. The results showed that the H_2S production in the MEC-AD reduced with an increase in applied voltage (94.8–98.2%) and the 1.6 V had the highest reduction. The AD of sulfur-containing compounds in sewage sludge produces H_2S (Krayzelova et al., 2015; Nguyen et al., 2018). The low H_2S content of the biogas produced from MEC-ADs at 1.3 V and 1.6 V indicated that the reduction of sulfur-containing compounds may have stopped or that H_2S may be converted further (Krayzelova et al., 2015; Nguyen et al., 2018; Chen et al., 2020).

The sulfate (SO_4^{2-}) concentration in the digestate was measured at the start and end of the test (Table 3). The result showed that the sulfate concentrations were below the detection limit in the start and end samples for MEC-ADs at 0 V and 0.8 V, and were 1.613 ± 0.111 and 8.017 ± 0.184 mg/L for the MEC-ADs at 1.3 V and 1.6 V, respectively (Table 3). The results showed that sulfate production occurred in the MEC-ADs at 1.3 V and 1.6 V. Elemental sulfur can be seen on the surface of digestate and electrode for the MEC-ADs at 1.3 V and 1.6 V (Fig. S4), which was confirmed by elemental analysis. No elemental sulfur was

Table 3
The hydrogen sulfide and sulfate concentration in the biogas and digestates.

Item	Start sample	End sample			
		0 V	0.8 V	1.3 V	1.6 V
H_2S in biogas (%)	–	0.302 ± 0.009	0.239 ± 0.008	0.005 ± 0.000	0.001 ± 0.000
SO_4^{2-} in digestate (mg/L)	BDL	BDL	BDL	1.613 ± 0.111	8.017 ± 0.184

BDL: Below the detection limit.

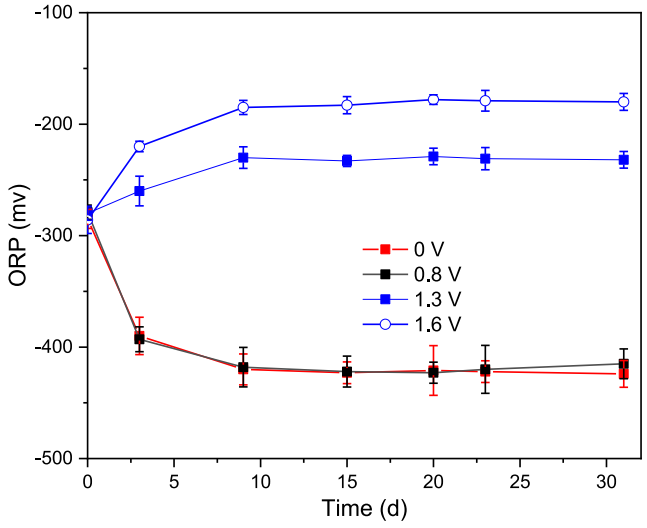


Fig. 2. Changes in the ORP of digestate in the test.

detected on the surface of digestate and electrode for the MEC-ADs at 0 V and 0.8 V (Fig. S4). The result suggested that elemental sulfur was produced in the MEC-ADs at 1.3 V and 1.6 V (Wang et al., 2020; Luo et al., 2020). Additionally, the amount of elemental sulfur accumulated on the surface of digestate and electrode for the MEC-AD at 1.3 V was much higher than that at 1.6 V (Fig. S4).

Therefore, the conversion of sulfur in the MEC-ADs includes the following processes:

1) Reduction of inorganic and organic sulfur-containing compounds, which produces H_2S (Mahdy et al., 2020; Nguyen and Khanal, 2018).
2) Oxidation of H_2S, which produces elemental sulfur (Eq. (1)). The oxygen produced in the anode diffuses into the cathode, providing electron acceptors for H_2S oxidation (Chen et al., 2020; Jung et al., 2022). SOB also could convert H_2S into produce elemental sulfur under a limited oxygen supply (Haosagul et al., 2020). Additionally, H_2S could also be directly oxidized into elemental sulfur on the anode (Eq. (2)) (Grgicak et al., 2008).

$$H_2S + {}^1\!/_2 O_2 \rightarrow S^0 + H_2O \quad (1)$$

$$H_2S \rightarrow {}^1\!/_2 S_2 + 2H^+ + 2e^- \quad (2)$$

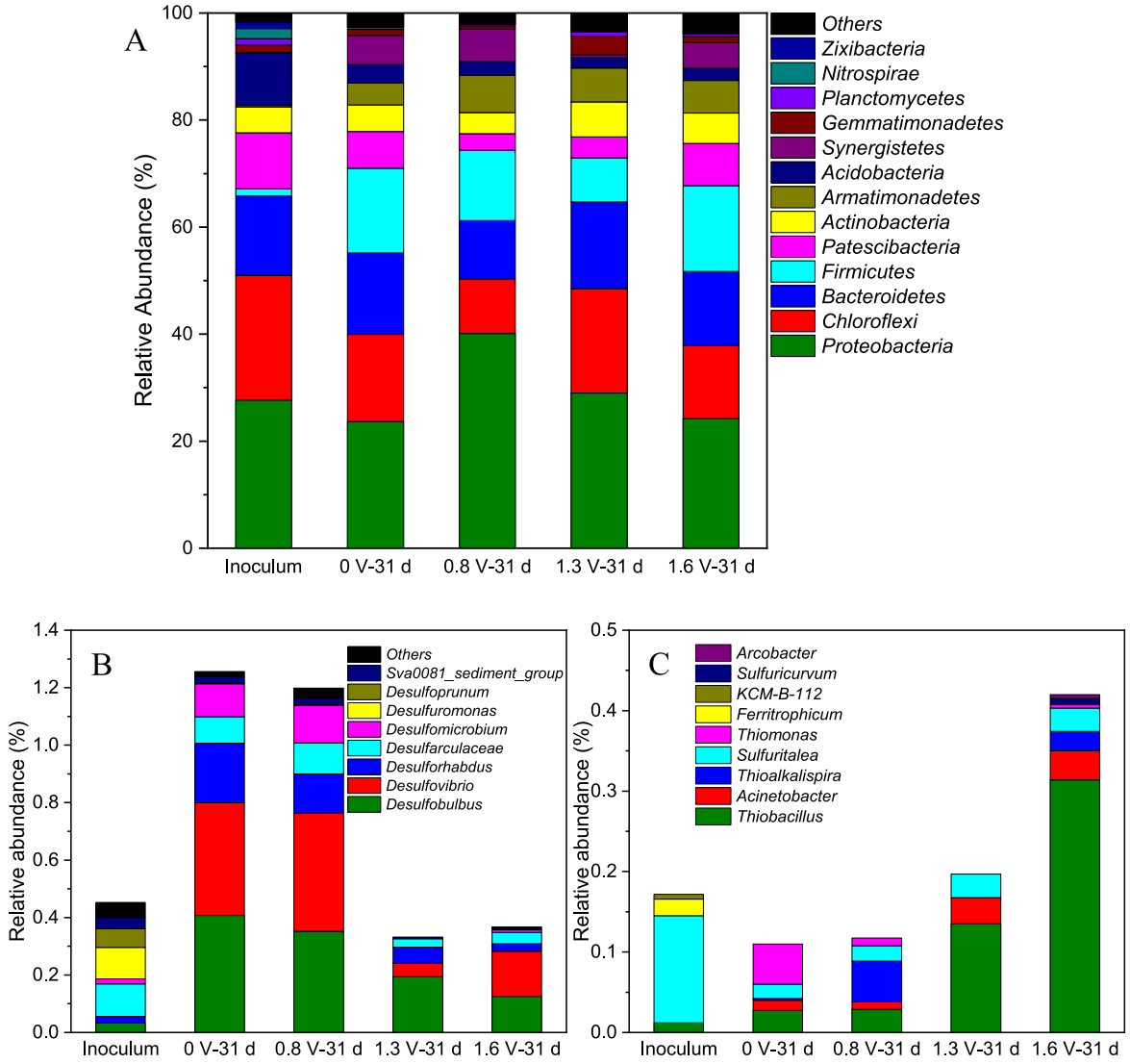


Fig. 3. Bacteria (phylum level) (A), SRB (B), and SOB (C) (genus level) communities of inoculum and digestate at the end of test.

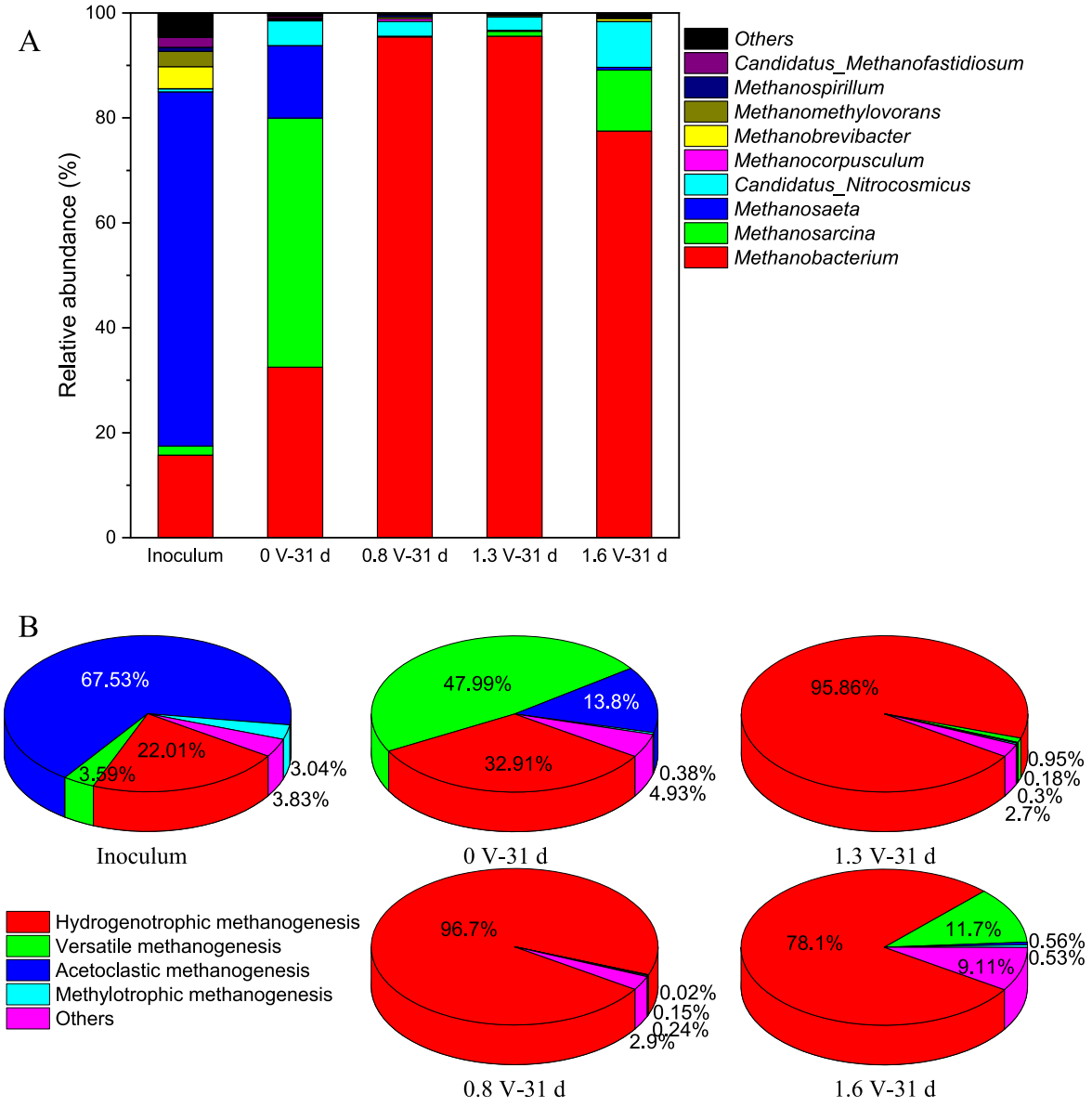


Fig. 4. Archaea communities at the genus level of inoculum and digestates (A) and methanogenesis pathways (B) at the end of the test.

3) Oxidation of elemental sulfur, which produces sulfate (Eq. (3)). In this process, the excess oxygen further oxidizes elemental sulfur to sulfate (Chen et al., 2020; Jung et al., 2022).

$$S^0 + H_2O + \frac{3}{2}O_2 \rightarrow SO_4^{2-} + 2H^+ \quad (3)$$

Only the first process occurred in the MEC-ADs at 0 V and 0.8 V, while all three processes occurred in those at 1.3 V and 1.6 V. More elemental sulfur oxidized to sulfate in the MEC-AD at 1.6 V than that at 1.3 V. Therefore, MEC-ADs at 1.3 V and 1.6 V reduced H_2S production as more H_2S was oxidized.

3.3. Changes in the ORP of digestates

The ORP is an important parameter for microaerobic conditions (Liu et al., 2013; Nguyen and Khanal, 2018; Wu et al., 2021; Muller et al., 2022), and the changes in the ORP of digestates in the digesters are summarized in Fig. 2. The results showed that the ORP of digestates in the MEC-ADs at 0 V and 0.8 V were similar, which increased from −285 ± −8 mv and −279 ± 6 mv to −424 ± −12 mv and −423 ± −13 mv during the test, respectively. The ORP of digestates in the MEC-ADs at 1.3 V and 1.6 V decreased from −280 ± −6 mv and −286 ± −12 mv to −232 ± −7 mv and −180 ± −8 mv, respectively. The results showed that the digesters at 0 V and 0.8 V were anaerobic conditions and those at 1.3 V and 1.6 V were microaerobic conditions (Nguyen and Khanal, 2018), that is, the applied voltages of 1.3 V and 1.6 V created microaerobic conditions for MEC-AD. The result indicated that oxide (oxygen) might produce in the MEC-AD at 1.3 V and 1.6 V, which increased the ORP of digestate and created a microaerobic condition for the MEC-AD (Chen et al., 2016; Tartakovsky et al., 2011). There are several reasons for why the increasing oxidation-reduction potential could enhance methanization in microaerobic condition: 1) facultative anaerobes (hydrolytic and acidifying bacteria) can rapidly consume the dissolved oxygen; 2) the microorganisms in the MEC-AD form microbial aggregates and methanogens are enveloped in the aggregates; 3) hydrolysis and acidification are enhanced in microaerobic condition, which

provide more substrates to methanogens; 4) some methanogens can tolerate oxygen exposure (Fu et al., 2023). Therefore the microaerobic conditions in the digesters at 1.3 V and 1.6 V was the reason for enhanced AD, reduced H_2S production, and increase in the sulfate concentration of digestates (Nguyen and Khanal, 2018; Muller et al., 2022). The ORP of digestate in the MEC-AD at 1.6 V was higher than that at 1.3 V (Fig. 2), indicating more oxygen production (Nguyen and Khanal, 2018), which was the reason for higher sulfate concentration of digestate in the digester at 1.6 V.

3.4. Microbial community

3.4.1. Bacterial community

At the phylum level, the most abundant bacterial populations were found to be *Proteobacteria* (23.64–40.13%), *Chloroflexi* (10.19–19.49%), *Bacteroidetes* (10.88–16.19%), and *Firmicutes* (8.27–7.91%) for the four digesters with different relative abundances at the end of the test (day 31) (Fig. 3-A and Table S1). The results suggested that the applied voltage affected the bacterial community of the digestate in MEC-AD. These results were similar to those of studies on MEC-AD (Chen et al., 2016) and microaeration-based AD with air or oxygen aeration (Zhuo et al., 2021). For example, Chen et al. (2016) found that the most abundant bacterial populations in the MEC-AD at 0–1.5 V applied voltages were *Proteobacteria*, *Firmicutes*, and *Bacteroidetes* for all reactors with different relative abundance. Zhou et al. (2021) found that *Proteobacteria* (29.40–34.89%), *Chloroflexi* (13.41–21.24%), and *Bacteroidetes* (6.57–12.17%) were the dominant phyla in the digestate during the AD of sludge with in situ microaeration. Among these bacteria, *Proteobacteria* is the main contributor to the anaerobic fermentation process, including the conversion of glucose, butyrate, propionate, and acetate, but not for methanogenesis (Zhang et al., 2021; Mahdy et al., 2020). *Chloroflexi* is responsible for the degradation of soluble microbial products and other complex substrates by producing hydrolytic enzymes, while *Bacteroidetes* and *Firmicutes* are known to include functional hydrolysers and fermenters in AD, and can be promoted by low-oxygen range-microaeration (Zhang et al., 2021).

SRB and SOB, two important bacteria involved in the conversion of sulfur (Rodrigues and Leão, 2020), were analyzed, and their relative abundances at the genus level are summarized in Fig. 3-B, C and Table S2. There were 8 SRB (relative abundance >0.03%) and 9 SOB in the digestates. The relative abundance of SRB in the digestates at 1.3 V and 1.6 V (0.33% and 0.36%, respectively) was lower than that at 0 V and 0.8 V (1.24% and 1.18%, respectively), with that at 0 V being the highest. The results for the SOB were contrary to those for the SRBs (Fig. 3-B and C). The relative abundance of SOB in the digestates increased with the applied voltage. The digestate at 1.6 V had the highest relative abundance of SOB (0.42%) and that at 0 V had the lowest (0.11%) (Fig. 3-C and Table S2). The results showed that a high applied voltage decreased the abundance of SRB and increased the abundance of SOB. The different abundances of SRB and SOB were consistent with the production of H_2S and sulfate (Fig. 1-C and Table 3). In the digestates, the dominant SRB and SOB were *Desulfobulbus*, *Desulfovibrio*, and *Thiobacillus* (Fig. 3-B and C). *Desulfobulbus* and *Desulfovibrio* are two common SRB found in AD with microaeration, which have been found to utilize ethanol, lactate, and hydrogen as electron donors and sulfate as electron acceptors to produce acetate (incompletely oxidizing SRB) (Liu et al., 2018; Mahdy et al., 2020; Wu et al., 2018). *Thiobacillus* is the most common SOB species found in the oxidation of sulfide (Krayzelova et al., 2015; Nguyen and Khanal, 2018).

3.4.2. Archaeal community

The archaeal communities of the inoculum and digestates for the four digesters at the end of the test (day 31) at the genus level are summarized in Fig. 4-A and Table S3. The results showed that the archaeal communities of the digestate were also affected by applied voltage. In the inoculum, the dominant archaea were *Methanosaeta* and *Methanobacterium*, with relative abundances of 67.53% and 22.01%, respectively. At the end of the test, significant changes were observed in the archaeal communities. In the MEC-AD at 0 V, the dominant archaea of digestate were *Methanosarcina* (47.99%) and *Methanobacterium* (32.91%). Unlike MEC-AD at 0 V, *Methanobacterium* became the dominant archaea (78.10–96.70%) in the three digesters with applied voltage at the end of the test, with MEC-AD at 1.6 V having the lowest abundance (Fig. 4-A).

There are four methanogenesis pathways: hydrogenotrophic, acetoclastic, methylotrophic, and versatile methanogenesis (Xiao et al., 2020). The methanogenesis pathways differed among the four digesters: hydrogenotrophic + acetoclastic methanogenesis for the digester at 0 V, and hydrogenotrophic methanogenesis for others (Fig. 4-B). These results suggest that MEC enhanced hydrogenotrophic methanogenesis, which is consistent with the findings of previous studies (Asztalos and Kim, 2015; Chen et al., 2016). However, the archaeal communities in this study differed from those of studies on microaeration-based AD with oxygen or air aeration (Fu et al., 2016; Zhou et al., 2021). For example, Zhou et al. (2021) found that in situ microaeration by air injection significantly enriched acetoclastic methanogens, rather than hydrogenotrophic methanogens. This difference may be due to the different microaeration methods used. Microaeration by air or oxygen injection only supplied oxygen or oxygen, nitrogen, and other trace gases, while microbial electrolysis at a high applied voltage (1.3 V and 1.6 V) used in this study simultaneously supplied O_2 and H_2 (Girotto et al., 2018; Tice and Kim, 2014). The hydrogen produced by electrolysis enhances the growth of hydrogenotrophic methanogens (*Methanobacterium*) (Hassanein et al., 2017; Chen et al., 2016). Additionally, the relative abundance of *Methanobacterium* in the digester at 1.6 V (78.10%) was lower than that in the digesters at 0.8 V and 1.3 V (95.43% and 95.58%, respectively) and that of *Methanosarcina* showed the opposite trend (Fig. 4-A), which may be due to the different oxygen sensitivity (Guo et al., 2021; Fu et al., 2016).

4. Conclusions

Applied voltages of 1.3 V and 1.6 V simultaneously reduced the H_2S production (by 94.8–98.2%) and enhanced methane production (by 12.70–57.02%) and organic matter removal (by 37.99–52.72% for VS) for the AD of sewage sludge. The two voltages electrolyzed water, increased the ORP of the digestate, and created microaerobic conditions. Sulfur reduction, H_2S oxidation, and elemental sulfur oxidation occurred in AD at 1.3 V and 1.6 V. The abundance of SOB increased and that of SRB decreased with an increase in the applied voltage. ADs at the two applied voltage enhanced the growth of *Methanobacterium*.

CRediT authorship contribution statement

Xiangyu Chen: Data curation, Writing – original draft, Writing – review & editing. **Benyi Xiao:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – original draft. **Xinyi Tang:** Data curation, Formal analysis, Investigation, Methodology. **Chunlin Bian:** Formal analysis, Writing – original draft. **Junxin Liu:** Conceptualization, Supervision, Resources, Writing – review & editing. **Lin Li:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Benyi Xiao reports financial support was provided by National Key Research and Development Program of China.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

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